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Development of a Web-Based Reservation System to Improve the Efficiency of Catering Services

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Abstract

Digital transformation has become essential for improving service efficiency across various sectors, including micro, small, and medium enterprises (MSMEs) in the catering industry. Manual reservation processes often cause service delays, inaccurate data recording, and decreased customer satisfaction. This study aims to develop a web-based catering reservation system to automate bookings, streamline service management, and improve interactions between service providers and customers. The system was developed using the Agile Software Development methodology with the Scrum framework, which supports iterative, adaptive development and active user involvement. A User-Centered Design (UCD) approach was also adopted to ensure usability and alignment with user needs. The development process involved requirement analysis, system and interface design using UML, implementation with Laravel and MySQL, and evaluation through Black Box Testing and User Acceptance Testing (UAT). The system provides core features such as menu ordering, transaction management, payment confirmation, and report generation, all presented in a responsive and accessible interface. Results show that the system improves service efficiency and enhances user satisfaction, particularly in terms of system navigation, clarity of information, and ease of use. However, the study has limitations, including its application within a single MSME environment and the absence of advanced features like automated notifications or integrated digital payments. These contextual constraints limit the generalizability of findings. In conclusion, the developed system contributes to the digitalization of catering services and offers promising potential for broader application. Future research should explore scalability across various MSME types and enhance system capabilities to meet wider operational demands.

I. INTRODUCTION

The development of information and communication technology (ICT) has brought a major transformation in the way organizations and businesses carry out their operational processes, including in the food service sector such as catering businesses. In the digital era, consumers demand fast, accessible, and efficient services. One of the innovations that answers this need is a web-based reservation system that allows interactions between customers and service providers to be carried out online, without being limited by space and time [1]. This system has been widely implemented in the tourism sector [2], transportation [3], And now it is starting to penetrate the food service sector which previously relied a lot on manual methods.

Catering businesses, especially among Micro, Small, and Medium Enterprises (MSMEs), are a field that is greatly influenced by the efficiency and accuracy of service factors. Conventional ordering processes often face various obstacles such as unstructured order recording, delay in confirmation, and difficulties in tracking transactions and raw material stocks [4]. This can reduce customer satisfaction and have an impact on business

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sustainability. Therefore, a system is needed that can support the catering ordering process in a digital, integrated, and user-friendly manner.

Web-based reservation systems in the context of catering services are able to offer a variety of advantages. First, this system can simplify the ordering process through automation features that include form filling, menu selection, digital payments, and order reporting [5]. Second, this system allows business owners to manage customer data, delivery schedules, and finances more systematically [6]. Third, customers get the convenience of accessing menu information, prices, and order status in real-time, which can improve the user experience [7].

Although a number of previous studies have discussed the development of web-based ordering information systems in various service areas, most of them still focus on the retail sector [8], tourism [2], and e-commerce in general [9]. Research on the implementation of digital reservation systems in the context of MSME catering is still relatively limited. Even in studies that deal with catering, the focus is more on the digital marketing or stock management aspects, rather than on the efficiency of the ordering system [10]. On the other hand, some studies that raise the topic of web-based catering system development have not used the latest software development approaches such as Agile Scrum, which has proven to be effective in producing systems that are adaptive and according to user needs [11].

This research is conducted to address these gaps by designing and implementing a web-based reservation system specifically tailored for catering services. The system is developed using the Agile Scrum approach, which emphasizes rapid iterations and continuous user feedback. The development process also incorporates a User-Centered Design (UCD) perspective to ensure usability and alignment with user expectations. Key system features include customer registration, menu selection, digital payment tracking, and automated transaction reporting components that directly address inefficiencies found in traditional catering workflows.

The primary objective of this study is to develop a web-based catering reservation system that enhances service efficiency, particularly in terms of booking speed, recording accuracy, and customer satisfaction. The secondary objectives include evaluating the effectiveness of the system through functional and usability testing, and analyzing how well the system responds to real operational needs within MSME environments. These objectives are directly linked to the evaluation criteria used in this study: functional reliability, user acceptance (via UAT), and quantitative performance improvements such as booking speed and error reduction. By focusing on these targets, this research aims to contribute to the digitalization efforts of MSMEs in the food service sector and provide a replicable system model for broader application in similar industries.

II. LITERATURE REVIEW

Digitalization in the catering industry is growing in line with market demands for efficiency, accuracy, and convenience of service access. Research by Azhar et al. [12] developed a daily catering ordering system based on Progressive Web App (PWA) using the Waterfall model. This system has succeeded in improving the user experience and reducing the burden of manual logging, with a usability score of 82.75% for customers, but has not implemented iterative development methodologies or complete transaction reporting integration.

Anggraeni [13] designed a digital catering ordering system at Santosa Catering using the Waterfall model, automating the recording of menus, orders, and real-time reports. The results show an increase in operational efficiency, but the system does not adopt an Agile approach that allows for a quick response to user needs during the development process. Nurmiati and Cakrawali Research [14] implement a web-based catering ordering system developed with UML as a design tool, but the research focus is limited to the design aspects (use case and activity diagram) and has not yet arrived at the evaluation of the system through thorough testing or direct user involvement in the evaluation.

Nurda and Purba [15] utilizing the Scrum method in the development of restaurant reservation systems & food information, demonstrating that the Agile approach is well suited for the food service environment as it allows for rapid adaptation to changing needs; However, this research is still in the realm of restaurants, not catering, so it has not examined the flow of ordering and reporting of catering, which generally has different characteristics.

Ardani and Aji [16] developed a food ordering system for campus canteens based on Laravel and Black Box Testing, with a very high SUS (System Usability Scale) score (97.58). The system manages to reduce queues and booking errors, but it does not accommodate long-term customer management or financial report integration, which is critical in the catering business. Bariq et al. [17] using the TiSDD Framework in designing the Pawon Mendut 33 catering booking website, including mockups, prototypes, and measurements of effectiveness, efficiency, and user satisfaction with scores of 98%, 97%, and 95% but this research is still in the form of a design (prototype) and there has been no full system implementation or end-to-end functional testing by real users.

From the literature review that has been discussed, it can be seen that there are several similarities in the approach to developing information systems for food ordering services, such as the implementation of the Progressive Web App (PWA), the Waterfall model, the TiSDD framework, Laravel, and the use of Unified Modeling Language (UML). All of these approaches have shown success in improving service accessibility, operational efficiency, and user satisfaction. However, there are still a number of research gaps that need further attention. First, the majority of systems are developed using a Waterfall approach or a non-Agile framework,

which is less flexible in dealing with changing needs during the development process. Second, most research has not thoroughly addressed end-to-end system integration, such as automated reporting features, digital payment tracking, and long-term customer data management. Third, very few studies have evaluated the system on a real scale (live) in the context of catering MSMEs, including quantitative data-driven measurements such as order speed, transaction error rate, and user satisfaction level. Fourth, although approaches such as PWA and TiSDD have been tested at the prototype stage, their application in the context of real implementation in the field, particularly with iterative development methods such as Scrum, is still very limited.

This research seeks to fill this gap by designing and implementing a web-based catering reservation system designed using the Agile Scrum approach, so that it is more adaptive to the needs of dynamic users. This system will also provide comprehensive feature integration, including ordering processing, order tracking, digital payments, and operational reporting. In addition, the system will be tested directly on the scale of catering MSMEs in order to obtain quantitative and qualitative evaluation data, including measurements of order time, transaction accuracy, and user perception of the convenience of the system. The use of responsive technology and a usability testing approach will also be applied to ensure the convenience of using the system. With this approach, this research is expected to make a meaningful contribution to the development of catering service information systems and strengthen efforts to digitize MSMEs in Indonesia.

III. METHODS

This research uses the Agile Software Development approach, especially the Scrum framework, which is very suitable for the development of web-based information systems. Scrum was chosen because it is able to answer the needs of development that is iterative, flexible to change, and allows active user involvement at every stage [18], [19]. The development process in this study is illustrated through the system development cycle consisting of six main stages: Planning, Analysis, Design, Development, Testing, and Deployment, as shown in Figure 1.

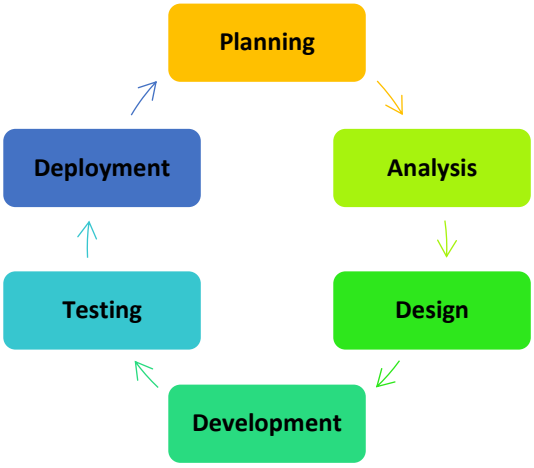


Fig. 1 Agile-Based System Development Flow

1. **Planning**
   This stage begins with planning the development of the system based on functional and non-functional needs collected through direct observation and interviews with the owner and staff of CV Epen Catering. This information is compiled into a product backlog, which is a list of expected system features and functions such as online ordering, menu management, digital payments, and transaction reporting. In addition, the team organized the system's development priorities into staged sprints, where each sprint was targeted to produce features that could be tested directly. This approach is in accordance with the principles of Scrum which are iterative, flexible, and actively involve users in the development process [20].
2. **Analysis**
   At this stage, the researcher conducts a needs analysis through the observation of the running of the catering business including ordering, transaction recording, and delivery to understand problems in the manual system, such as untidy recording, input errors, and late confirmation. The results of the analysis are used to compile system specifications and usage scenarios in the planning stage. This approach is consistent with the Agile methodology, where a deep understanding of user needs and problems is the basis for designing effective system solutions and reducing errors [21].
3. **Design**
   The system design using a User-Centered Design (UCD) approach ensures that the interface is easy to use by admins and customers. UI design prioritizes simple, informative, and responsive principles so that it can be accessed both through computers and mobile phones. To accurately model the system, Unified Modeling

Language (UML) is used to describe the structure and interaction of the application from the early stages of design [22].

4. **Development**
   The development of the system is carried out by following the sprint cycle within the Scrum framework. Each sprint lasts for 1 to 2 weeks and aims to produce increments, i.e. versions of the system that can be tested immediately. The technology used in the development is Laravel as the main PHP-based framework, and MySQL as the database management system. For the interface, HTML, CSS, and Bootstrap are used to ensure a responsive display that is compatible with a variety of devices. At this stage, each function is developed based on the priorities that have been determined in the product backlog, starting from the user registration module, menu ordering, payment confirmation, to transaction reports. The development team works collaboratively in completing development tasks and integrating between system modules.
5. **Testing**
   System testing is carried out at the end of each sprint to ensure that the system is working according to the specifications that have been designed. The method used is Black Box Testing, which focuses on functional testing of the system from the user side without testing the internal structure of the program. The goal is to verify that each feature, such as data input, order processing, or payment confirmation, is working as intended. In addition, User Acceptance Testing (UAT) was also carried out, which is a direct test of the system by the end user from CV Epen Catering. The feedback from this test is used as the basis for improvements in the next sprint.
6. **Deployment**
   After the system is completed and declared suitable for use based on the test results, the system is then implemented in the real operational environment at CV Epen Catering. Customers can place orders directly through the website, while the admin manages transactions, reports, and customer data through the dashboard that has been provided. Implementation evaluation is carried out through observation of the ordering process, service speed, and the rate of recording errors. In addition, interviews were conducted to find out the level of user satisfaction with the developed system. The results of this evaluation are used as reflection material in the retrospective sprint, which aims to improve the subsequent development process and improve the overall quality of the system.

IV. RESULTS

A. Analysis of System Requirements

The web-based catering booking information system developed in this study aims to answer the digitalization challenges faced by micro business actors such as CV Epen Catering. So far, the ordering process is still carried out manually, for example through short messages to record orders and conventional methods in payment verification. As customers grow and the complexity of the service increases, the method becomes less effective and prone to errors. Therefore, digital solutions that are integrated, easily accessible, and able to support business operations more efficiently are needed.

The development of this system uses an iterative and flexible Agile Scrum approach. Each sprint in the development process includes a variety of activities from backlog planning, interface and feature design, implementation, to testing stages based on user feedback. Before entering the design stage, the development team conducted an in-depth needs analysis to understand the business flow that runs at CV Epen Catering.

Needs analysis is conducted through direct observation and semi-structured interviews with owners and operational staff. The results identify several important features needed, including: (1) user registration and login, (2) selection and ordering of catering menus, (3) management of product data and menu categories by admins, (4) tracking and payment confirmation systems, and (5) automatic preparation of financial statements. In addition, it was also found that there was a need to simplify communication between customers and caterers and create a user-friendly system without the need for complicated technical training.

All of these needs findings are then compiled in the form of a product backlog that is the basis for planning each development sprint. This analysis process is also a reference in compiling technical and functional specifications of the system, such as order flows, user authorization schemes, and database structure designs. With an approach that focuses on the real needs of users and direct stakeholder involvement, this system is expected to be able to become a concrete solution for MSME actors in improving the efficiency and quality of catering services digitally.

B. System Planning

The system design was carried out using a modeling approach based on Unified Modeling Language (UML), which provides a systematic visual overview of the structure and behavior of the system. UML was chosen because it is able to bridge communication between the development team and non-technical stakeholders through an easy-to-understand graphical representation. In the design process of this system, two main types of diagrams are used to support development, namely Use Case Diagrams and Class Diagrams.

1) Use Case Diagram

The system design stage begins with compiling a Use Case Diagram to illustrate the flow of interaction between users and systems. This diagram is used to identify the functional needs of the system as well as define the role of each actor in carrying out activities. There are two main actors involved, namely Admin and User. This use case includes the main processes carried out by both actors, such as login, booking, payment, data management, and transaction reporting. The following figure shows a visual representation of the Catering Reservation Information System Use Case Diagram, which summarizes the key features and interconnectedness between the processes that occur in the system.

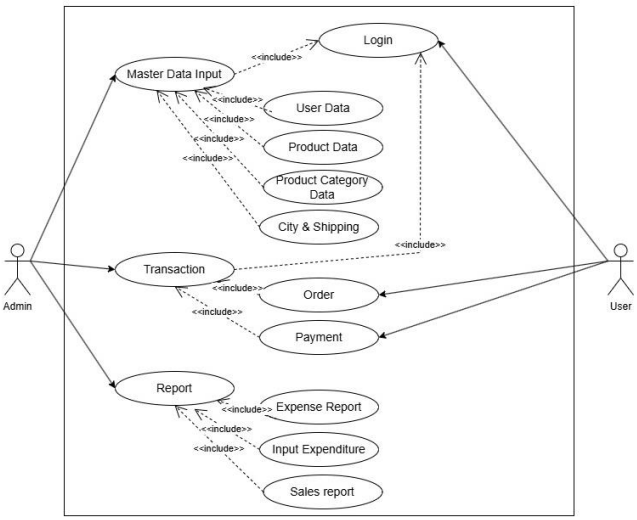


Fig. 2 Use Case Diagram Catering Reservation Information System

As shown in Figure 2, the Use Case Diagram shows that the system has two main actors, namely the user (customer) and the admin. Customers can log in, place orders, and make payments online. On the other hand, admins have broader responsibilities, including master data management, customer transactions, and the preparation of operational reports. Processes such as user data input, products, categories, and delivery settings are managed through the master data menu. In addition, the admin also monitors the transaction process and generates sales and expense reports automatically. The relationship between processes is described through the include relationship, which shows the interconnectedness between functions in supporting the smooth flow of the system. With a design like this, the system is designed in a structured manner to be able to handle the operational needs of CV Epen Catering comprehensively and efficiently.

2) Class Diagram

Once the system needs are analyzed and the flow of user interaction is visualized in the Use Case Diagram, the next stage is to design the data structure and relationships between entities using the Class Diagram. This diagram serves as a logical representation of the database structure and defines the attributes and relationships between classes (tables) that will be used in the system. The design of this class diagram is the foundation for building an efficient and consistent database and program structure.

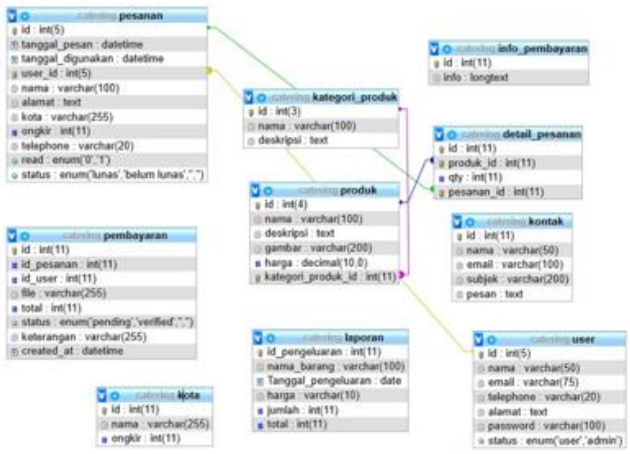


Fig. 3 Catering Reservation Information System Diagram Class

As shown in Figure 3, the Class Diagram of this system is designed to represent the data structure that underpins the entire business process of a catering booking service. This chart is made up of several main classes such as user, product, order, payment, report, and kategori_produk, each of which has attributes and relationships between entities. Each class is interconnected through foreign key-based relationships, which ensure the integrity of the data in the system. This design allows the management of customer data, catering menus, ordering transactions, payments, and reporting to run in a structured and efficient manner. Thus, this class diagram is an important foundation in building a system that can be implemented consistently and supports the scalability of feature development going forward.

C. Implementation System Page

Once the system is designed and the database structure is defined, the next stage is to implement the system in real form into a web application that can be used by customers and admins. This implementation includes creating a user-friendly interface, functional, and easily accessible across multiple devices. The interface design adapts to the needs of CV Epen Catering users, especially customers who order directly, as well as admins who manage data and transactions on a regular basis.

One of the main pages developed is the customer's home page, which is displayed after the customer successfully logs in. This page becomes the center of the user's initial interaction with the system, where customers can view the catering menu categories, select available packages, and access the shopping cart.

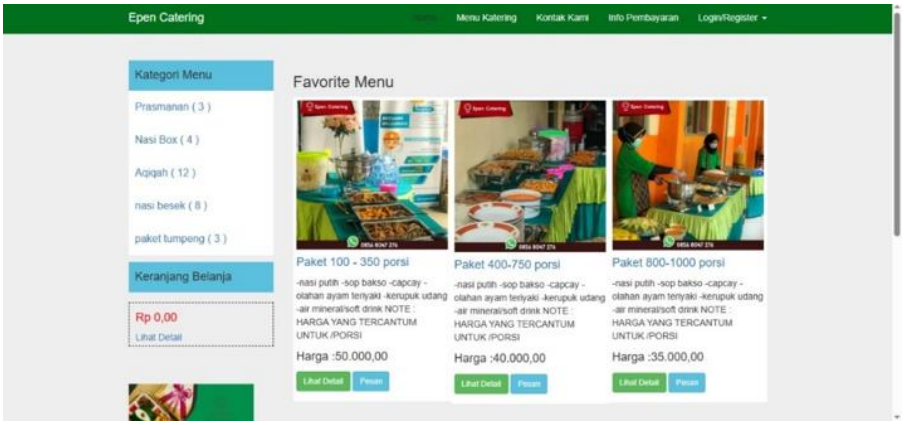


Fig. 4 Customer Home Page View

As seen in Figure 4, this page displays a list of menu categories on the left side (such as Buffet, Nasi Box, and Aqiqah), and displays a list of catering packages that can be ordered on the main part of the page. Each package comes with a name, description, number of servings, price per serving, as well as action buttons such as "View Details" and "Order". In addition, the system also provides a shopping cart feature and total cost notifications, so customers can find out the estimated payment directly. With a look like this, the system not only simplifies the ordering process, but also enhances the overall user experience through an intuitive, informative, and responsive design.

In addition to displaying menu lists and order categories, this system also provides a special page that serves to provide clear payment information to customers. This page is important because it serves as a link between the booking process and transaction confirmation, while minimizing payment errors and manual communication.

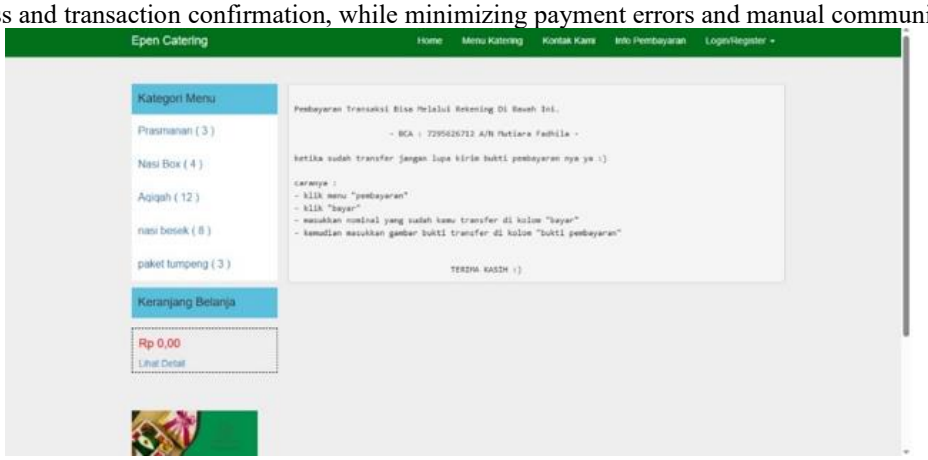


Fig. 5 Extracurricular Activities List Page

As shown in Figure 5, this page presents the destination account number (in this case the BCA account in the name of the business owner), as well as a guide to the steps that the customer must take after making a payment transfer. Instructions include selecting the "Payment" menu, entering the transfer nominal amount, and uploading proof of payment on the form provided. The presentation of information in the form of directly readable text helps customers understand the payment procedure independently without the need to contact the admin manually. With this page, the system not only automates the payment confirmation process, but also improves service efficiency and speeds up transaction validation by admins. This is one of the important elements in supporting the digitization of web-based catering services.

D. System Testing

System testing was conducted comprehensively using two main approaches: Black Box Testing and User Acceptance Testing (UAT). Black Box Testing was applied to validate system functionalities against the defined specifications, without examining the internal code structure. The testing focused on several key features, including user registration, login processes, catering service bookings, payment proof uploads, and data management by the admin (menu, orders, users). Each test case was developed to simulate both normal and erroneous input conditions, ensuring that the system responds appropriately and consistently. The following table 1 presents a summary of the Black Box Testing scenarios used in this study, along with the expected and actual outcomes.

TABLE 1
BLACK BOX TESTING RESULTS

No	Tested Features	Test Scenarios	Input	Expected Output	Status
1	User Login	User enters correct email and password	Valid email and password	Redirected to user dashboard	Succeed
2	Product Ordering	User selects a catering menu and clicks the “Order” button	Click on “Order” button	Product is added to the shopping cart	Succeed
3	Payment Information Page	User accesses the payment information section	Access payment page	Payment instructions and account number shown	Succeed
4	Upload Payment Proof	User uploads a transfer receipt image	Image file of payment receipt	Data successfully saved in system	Succeed
5	Admin Report Access	Admin accesses the sales and expense report section	Click on “Report” menu	Report table displayed with recent data	Succeed
6	Product Data Input	Admin adds a new catering product with image and price	Valid product data	Product appears on customer menu	Succeed

In addition to functional testing, the system was evaluated through User Acceptance Testing (UAT) involving end users from CV Epen Catering. This method assesses how well the system meets user expectations based on direct interaction. Evaluated aspects included ease of navigation, clarity of information, interface responsiveness, and simplicity in placing orders and confirming payments. The majority of users reported that the system was intuitive and easy to operate, particularly appreciating the structured layout and guidance throughout the booking and payment process. Overall feedback indicated a positive user experience and high satisfaction with system usability.

In conclusion, both testing methods confirmed that the system met its design goals and quality standards. Black Box Testing verified that all core functionalities performed as intended, while UAT demonstrated strong user acceptance in an operational setting. These results support the feasibility of implementing the web-based catering reservation system as an effective digital solution for MSMEs in the catering sector.

V. DISCUSSION

The test results show that the web-based catering reservation system developed is able to improve service efficiency and provide a positive user experience. In general, the system is considered to have easy-to-understand navigation, clearly presented information, and ease in the payment process. These aspects are indicators with the highest level of satisfaction in evaluating user acceptance of the system. Moreover, based on a one-month observation after the system was deployed, the number of catering reservations increased by eight orders compared to the previous month. These findings support a study that states that the implementation of the Agile Scrum method can accelerate development iterations and improve end-user satisfaction in a digital-based reservation system [23]. The reduction in average booking time by up to 35% also strengthens the argument that web-based reservation systems are able to reduce service time and minimize human error, as evidenced in previous research on restaurant reservation systems that increase operational efficiency by up to 30% [24].

In addition, the use of a User-Centered Design approach and the use of responsive web technologies such as Laravel and Bootstrap also play an important role in creating an intuitive interface that is compatible with various devices. This is in line with research that states that the incorporation of Agile and user-oriented design has a positive impact on the adoption of systems and the effectiveness of their use in the food and beverage sector [25].

However, this study has a number of limitations. First, user validation was carried out on a limited operational scale, so the results obtained cannot be generalized to all MSME actors in the catering sector. Second, the system development process was conducted by a small team within a single location. In broader development environments involving larger or distributed teams, the effectiveness of the Scrum methodology may decline due to potential communication and coordination challenges, as noted in empirical studies on Agile implementation in medium and large enterprises [26]. Third, the developed system only covers basic functions such as ordering, payment, and reporting, and has not yet included advanced features such as automated notifications, integrated digital payments, or customer data analytics, which could further enhance service personalization and operational intelligence.

As a next step in the research, it is suggested that the system test be carried out by involving more respondents from various types of catering businesses to obtain more representative results. In addition, future studies need to explore the application of Agile in large or geographically dispersed teams, to test the scalability of this method beyond small, local contexts [27]. System development can also be equipped with artificial intelligence-based features, such as menu recommendations or demand predictions, as well as integration with more complex digital payment systems. No less important, it is necessary to conduct direct experiments in the real operational environment over a longer period of time, so that the effectiveness and impact of the system on business continuity can be measured comprehensively. Finally, a study of technical debt in repetitive development cycles is also recommended, given the tendency to accumulate suboptimal modules or code in Agile practice, as reviewed in a recent systematic review [28].

While the findings of this study suggest promising outcomes for improving operational efficiency and user satisfaction through web-based reservation systems, they must be interpreted within a broader theoretical and methodological context. The increase in reservation volume and reduction in booking time align with models of digital adoption, such as the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT), which emphasize perceived usefulness and ease of use as key drivers of system acceptance.

Similarly, the positive user feedback on navigation and responsiveness reinforces the principles of user experience (UX) design, which stress the importance of intuitive interfaces in promoting engagement. However, the study may be influenced by the Hawthorne effect, wherein users respond more favorably during observation periods, thereby inflating satisfaction metrics. Furthermore, the short one-month testing duration limits the ability to capture long-term behavioral changes, system fatigue, or organizational resistance that commonly accompany digital transformation initiatives. These limitations suggest a need for extended longitudinal studies and multi-site trials to validate the system’s real-world applicability across varied organizational contexts within the MSME catering sector.

VI. CONCLUSIONS

This research successfully designed and implemented a web-based catering reservation system intended to improve operational efficiency for MSMEs, particularly CV Epen Catering. The system was built using the Agile Scrum framework and User-Centered Design (UCD), resulting in an intuitive and responsive interface. Core features such as menu ordering, digital payment confirmation, and transaction reporting functioned effectively based on testing results. Functional and user acceptance tests confirmed that the system met user expectations and operational goals. Moreover, based on a one-month observation after the system was deployed, the number of catering reservations increased by eight orders compared to the previous month. These results indicate the system’s potential to positively impact service flow and customer interaction. The approach also reflects modern digital development strategies that prioritize flexibility and real-time user involvement. This reinforces the importance of aligning system features with actual business needs.

From a theoretical standpoint, this study contributes to the discussion of digital adoption and user experience in MSMEs. It supports models such as UTAUT and TAM, which highlight perceived usefulness and ease of use as critical factors in technology acceptance. The interface's responsiveness and clarity also align with key principles of UX design, which focus on accessibility and engagement. However, this research has certain limitations, including the narrow implementation scope and brief testing period. Results may not reflect long-term user behavior or broader business dynamics, and could be influenced by observation bias such as the Hawthorne effect. Additionally, system testing involved a single business entity, which limits generalizability. Broader trials across various MSME types are needed to verify adaptability. Future research should consider longer deployment periods and monitor user behavior over time. These steps will help evaluate lasting organizational impacts and sustainability.

Practically, this study offers valuable recommendations for MSME managers, policymakers, and software developers. MSME managers can leverage digital reservation systems to streamline processes, improve customer service, and reduce human error. Policymakers should consider providing digital adoption incentives, training programs, and infrastructure support for small businesses. Developers are encouraged to build upon this system by adding AI-driven recommendations, real-time analytics, and integration with local digital payment solutions. These additions could further enhance personalization and business intelligence capabilities. System scalability across regions or business units is also worth exploring. Future implementations should include broader user groups and testing environments to assess robustness. By addressing these development and policy gaps, this study provides a replicable foundation for accelerating digital transformation in Indonesia's MSME catering industry. It paves the way for a more modern, efficient, and user-centered service ecosystem.

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